

Question Paper

Exam Date & Time: 26-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Fundamentals of Machine Learning [CSE_3127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) You are given a dataset with overlapping classes where hard-margin SVM fails to converge. After applying soft-margin SVM with different C values ($C = 0.01$, $C = 1$, $C = 100$), you observe varying performance on training and test sets. (5)
- A) (i) With the help of neat diagrams, illustrate how the margin width, support vectors, and misclassified points. Then, analyze how the choice of C affects the number of support vectors, margin width, and classification errors.
(ii) Using the bias-variance decomposition framework, analyse why extremely small and extremely large C values both lead to poor generalization. Support your analysis with reasoning about underfitting and overfitting.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) With the help of a diagram, explain the Stacking ensemble approach along with its major steps. Analyse how the meta-model in stacking improves prediction accuracy by combining strong learners. Explain the working of Bagging (Bootstrap Aggregation) with the help of a neat diagram and the main steps involved. (3)
Analyse how Bagging reduces variance and improves model stability.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) **For the matrix** (2)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

apply the SVD procedure to find the singular values of A.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Evaluate how pre-pruning and post-pruning techniques affect the generalization performance of a decision tree. Using the principle of cost-complexity pruning, analyse the trade-off between model simplicity and accuracy. (4)
- A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) List the steps of the XGBoost algorithm. Analyse how learning from residuals differs from AdaBoost's weight-based approach. (3)

- C) Analyse the model complexity versus error curve and explain how changes in model complexity influence training and test errors. Identify the region corresponding to underfitting, overfitting, and the optimal generalization point. (3)

- 3) You are given the following training data: (4)

A)

PetalSize	Color	FlowerType
Small	Red	Rose
Medium	Yellow	Tulip
Large	Yellow	Sunflower
Small	Pink	Rose
Medium	Yellow	Tulip
Large	Orange	Sunflower

Predict the flower type for a new flower with PetalSize = Medium and Color = Yellow. Apply the Naïve Bayes classifier to find the most probable FlowerType.

- B) In noisy or non-separable datasets, the Perceptron learning algorithm keeps updating the hyperplane indefinitely. Analyze why this happens and how introducing a margin (e.g. in case of SVM) could mitigate the issue. (3)

- C) **A logistic regression model predicts whether an employee will resign within 6 months:** (3)

$$P(\text{Resign}) = \frac{1}{1 + e^{-(-4 + 0.08x_1 + 0.6x_2)}}$$

where

x_1 = Workload (hours/week)

x_2 = Job Satisfaction (1 = Low, 0 = High)

For an employee working 45 hours/week with low satisfaction,

1. Calculate the probability of resignation.
2. State the classification if threshold = 0.55.
3. If workload reduces to 40 hours/week, re-evaluate the prediction.

- 4) You are classifying drones based on their Speed (km/h) and Battery Endurance (hours) to predict the mission type they are best suited for. (5)

A)

ID	Speed (x)	Endurance (y)	Mission
D1	2	2	Survey
D2	6	5	Delivery
D3	20	20	Delivery
D4	9	2	Survey
D5	12	2	Survey
D6	8	7	Delivery

Two test drones are observed:

1. $T_1 = (7.5, 5.0)$
2. $T_2 = (9.0, 3.0)$

Apply the k-NN classification algorithm with Euclidean distance and $k = 3$ to predict the mission type for both test drones. Show all distance calculations and the final class for T_1 and T_2 .

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Analyse how problem-transformation methods (such as Binary Relevance or Classifier Chains) differ in capturing inter-label dependencies, and the structural reason behind their performance differences. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Analyse how the Linear Kernel, Polynomial Kernel, and RBF Kernel differ in terms of their ability to handle non-linear decision boundaries. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) Analyse how introducing a time-decay function in Timed PageRank modifies the random-surfer model of traditional PageRank, and examine the key steps of the Timed PageRank algorithm. (4)

A)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- B) Analyse how Content-Based Filtering builds user profiles using item feature vectors such as TF-IDF representations. (3)
Explain Content-Based Filtering and how this approach helps mitigate the cold-start problem for new users but struggles with over-specialization.

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- C) Analyse how Co-Citation helps in identifying the similarity between two documents. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3127:Fundamentals of Machine Learning

#	Course Outcome	Description	Marks
1	1	CO 1	2
2	2	CO 2	4
3	3	CO 3	12
4	4	CO 4	11
5	5	CO 5	11
6	6	CO 6	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3127:Fundamentals of Machine Learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	14(28%)
4	Analysis	11	36(72%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)